Introduction to Reinforcement Learning (EE675): Assignment 2

Q. 1: Consider the same predator-prey game in an $N \times N$ grid as discussed in Assignment 1, where $N = 4$.

(a) Represent the policy function as a neural network of appropriate dimensions. Please use the PyTorch module to design neural components. Explain your design choices in a PDF file that you will submit alongside the code files.

(b) Write a gradient_estimate function that obtains an unbiased estimate of the policy gradient using the simulator function. Explain the process in the PDF file. Explain which PyTorch in-built method is used to obtain the score function (i.e., gradient of the log of the policy function).

(c) Write a simple_SGA function that updates the neural policy parameters using the simple stochastic gradient ascent step discussed in the class. Mention your choice of learning rates. Plot the learning curve of your algorithm (i.e., total reward obtained as a function of number of iterations) in the PDF file.

(d) Familiarize yourself with other in-built optimizers of the PyTorch module. Plot the learning curve of the policy gradient algorithm in the PDF file if the Adam optimizer is used instead of simple_SGA.

(e) Write a readme.txt file to clearly explain how to run the code. Put the readme.txt, PDF, and code files (.py) in a single folder, compress it, and upload it as a $.\text{zip}$ file.